

Lecture 2: Perspectives on Diffusion

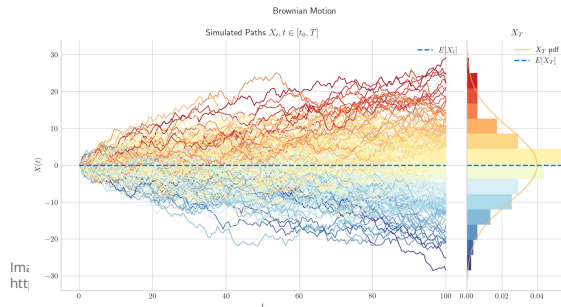
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Recap: What are diffusion models?

Inspired by diffusion processes
(Brownian motion)



Forward process adds noise to data

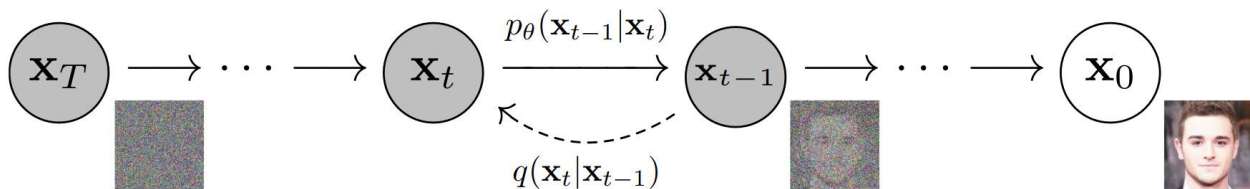


Image credit: Denoising Diffusion Probabilistic Models <https://arxiv.org/pdf/2006.11239>

Diffusion models learn the reverse process to recover data from noise

Perspectives and Interpretations

1. **Notation:** Continuous vs Discrete Diffusion Processes
2. **Deterministic sampling:** Transport Equation and Flow-ODE
3. Discrete sampler: DDIM
4. **Stochastic sampling:** SDEs and Fokker-Planck
5. **Converting between model representations:**
x0-prediction, epsilon-prediction, v-prediction, and score-prediction
6. Discrete sampler: DDPM



Continuous vs Discrete Diffusion Processes

Discrete Processes

Timesteps

$$t \in \{1, \dots, T\}$$

Noise levels

$$\sigma_t \in \{\sigma_1, \dots, \sigma_T\}$$

Diffused samples

$$x_t \in \{x_1, \dots, x_T\}$$

$$x_t \stackrel{\text{dist}}{=} x_0 + \sigma_t \epsilon,$$
$$\epsilon \sim \mathcal{N}(0, I)$$

Continuous Processes

$$t \in [0, 1]$$

$$\sigma : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$$

$$x : [0, 1] \rightarrow \mathbb{R}^d$$

$$x(t) \stackrel{\text{dist}}{=} x(0) + \sigma(t) \epsilon,$$
$$\epsilon \sim \mathcal{N}(0, I),$$
$$\sigma(0) = 0$$



Continuous vs Discrete Diffusion Processes

Discrete Processes

$$\begin{aligned}x_1 &= x(t_1), & \sigma_1 &= \sigma(t_1), \\ \dots, & & \dots, \\ x_T &= x(t_T) & \sigma_T &= \sigma(t_T)\end{aligned}$$



$$\begin{aligned}x_t &\stackrel{\text{dist}}{=} x_0 + \sigma_t \epsilon, \\ \epsilon &\sim \mathcal{N}(0, I)\end{aligned}$$



Discretize at
 $\{t_1, \dots, t_T\}$

**We will use
continuous processes
for all our math!**

Continuous Processes

$$t \in [0, 1]$$

$$\sigma : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$$

$$x : [0, 1] \rightarrow \mathbb{R}^d$$

$$x(t) \stackrel{\text{dist}}{=} x(0) + \sigma(t)\epsilon,$$

$$\epsilon \sim \mathcal{N}(0, I),$$

$$\sigma(0) = 0$$



Continuous Diffusion Processes

To simplify notation, from here on we will use **continuous processes** everywhere but may use **subscripts** for the “time” of the diffusion process when convenient, i.e.

$$\sigma_t := \sigma(t) \in \mathbb{R}_{\geq 0}$$

$$x_t := x(t) \in \mathbb{R}^d$$

$$x_t \stackrel{\text{dist}}{=} x_0 + \sigma_t \epsilon,$$

$$\epsilon \sim \mathcal{N}(0, I),$$

$$\sigma_0 = 0$$



What is a diffusion process?

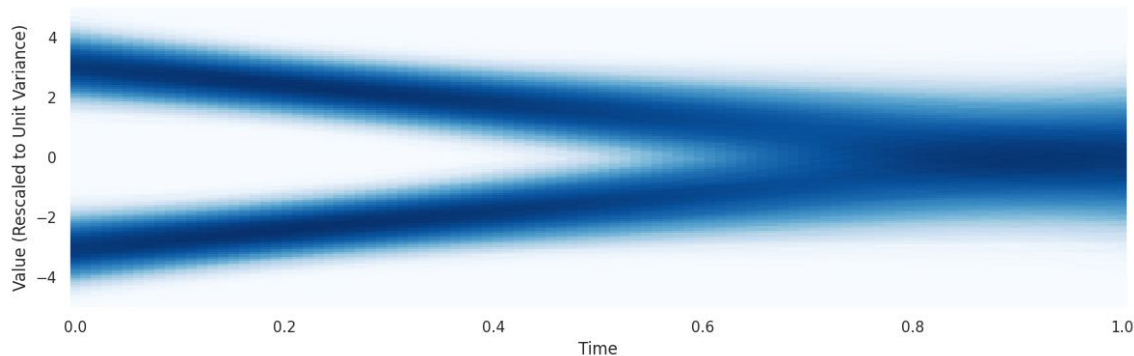
$$x_t \stackrel{\text{dist}}{=} x_0 + \sigma_t \epsilon,$$

$$\epsilon \sim \mathcal{N}(0, I),$$

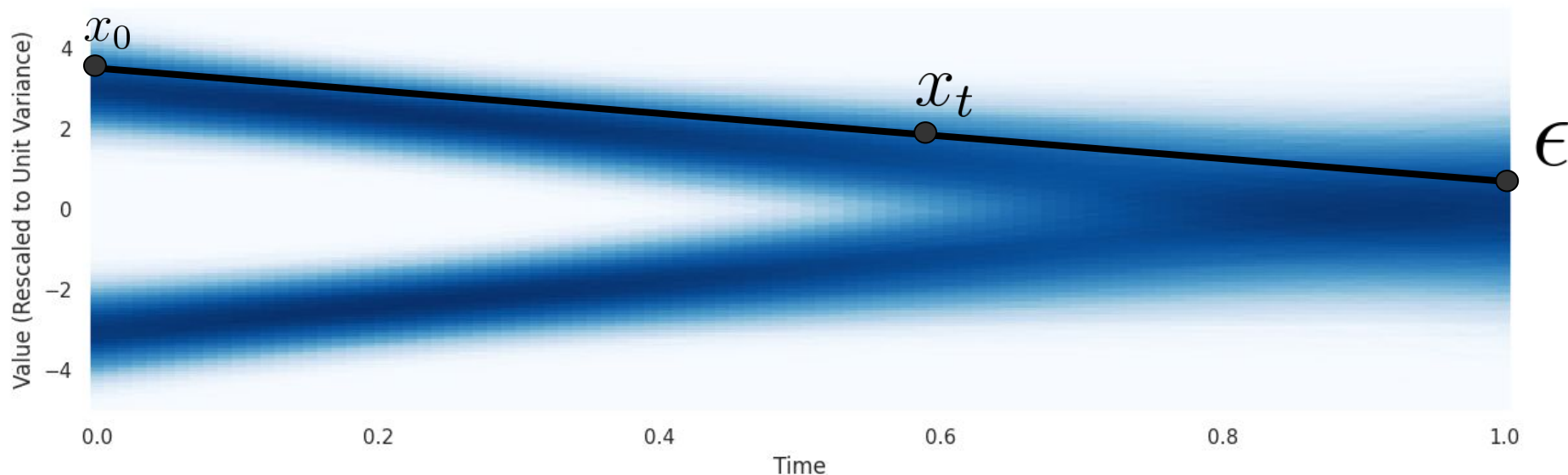
$$\sigma_0 = 0$$

We define only the **marginal** distribution $p(x_t)$ for each t .

The **joint** distribution $p(x_s, x_t)$ is dependent on the **sampling method or process** that we use.



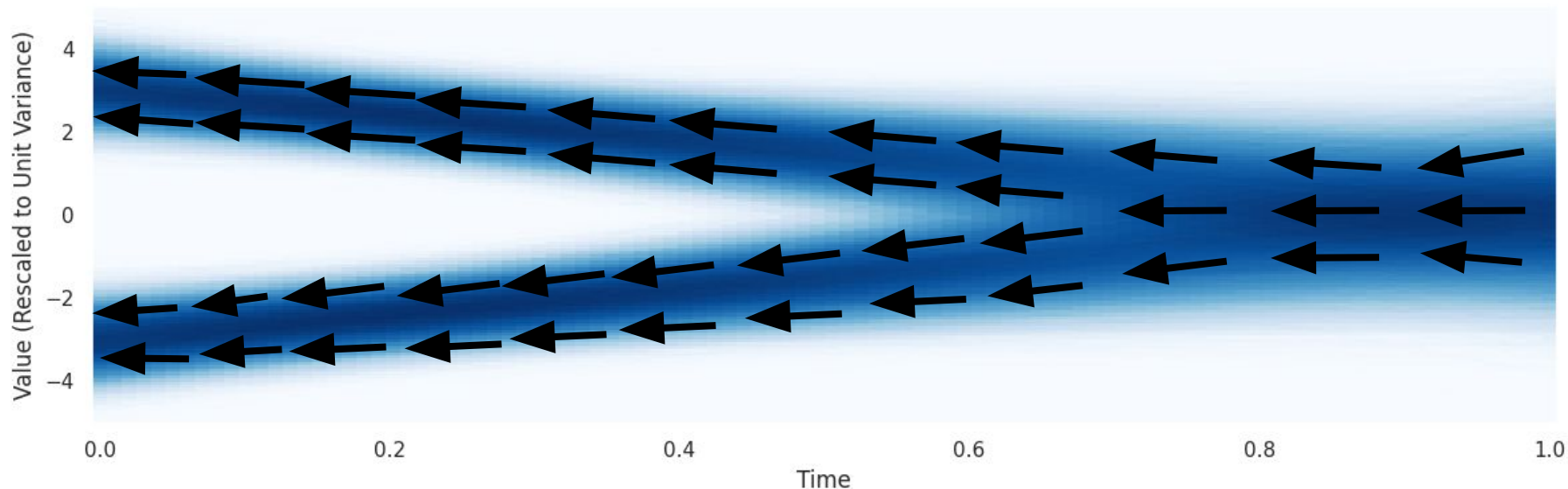
Stochastic Interpolant: Sampling $p(x_t)$



$$x_0 \sim p^*(x), \quad \epsilon \sim \mathcal{N}(0, I), \quad x_t = x_0 + \sigma_t \epsilon$$

Useful for generating samples from $p(x_t)$, but requires samples from data distribution!

Flow-ODE: Deterministic Sampling



What is the flow $v(x, t)$ such that $\dot{x}_t = v(x_t, t)$, $x_1 \sim \mathcal{N}(0, \sigma_1^2 I)$ satisfies $x_t \stackrel{\text{dist}}{=} x_0 + \sigma_t \epsilon$ provided $\sigma_1 \gg 0$?

Transport Equation

Continuity Equation: A time-varying family of densities $p_t(x)$ and a flow $v(x, t)$ are consistent with the ODE $\dot{x}_t = v(x_t, t)$ iff they satisfy:

$$\frac{\partial p_t(x)}{\partial t} + \nabla \cdot (v(x, t)p_t(x)) = 0$$

We need to find a solution $\mathbf{v}$ for the $p(x_t)$ distributed according to $x_t = x_0 + \sigma_t \epsilon$.

Turns out the solution is:

$$v(x_t, t) = \dot{\sigma}(t) \mathbb{E}[\epsilon \mid x_t]$$

where $x_t = x_0 + \sigma_t \epsilon$

Recall: DDIM Sampler from Last Lecture

Continuous ODE-Flow

$$\dot{x}_t = v(x_t, t)$$

$$v(x_t, t) = \dot{\sigma}(t) \mathbb{E}[\epsilon \mid x_t]$$

where $x_t = x_0 + \sigma_t \epsilon$

Discrete DDIM Sampler

Algorithm 1 DDIM sampler

for $t = N, \dots, 1$ **do**

$x_{t-1} \leftarrow x_t + (\sigma_{t-1} - \sigma_t) \epsilon_\theta(x_t, \sigma_t)$

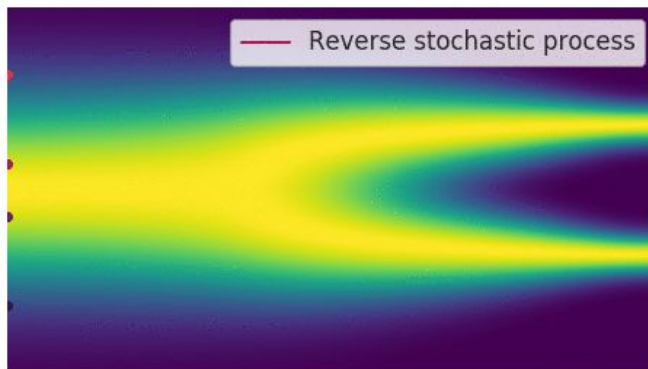
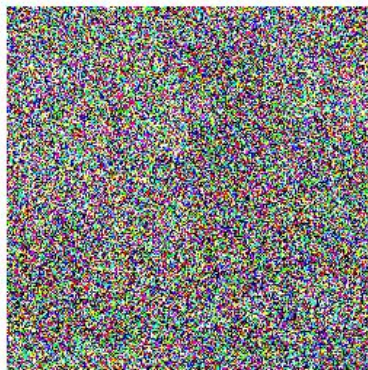
return x_0

The DDIM sampling algorithm is just a Euler discretization of the continuous ODE flow!

Stochastic Reverse Processes

There is another way to traverse the **same marginal distributions** based on **random walks** and **stochastic processes**.

These require more steps, but ultimately produce higher quality results!



Random Walks: Stochastic Differential Equations

$$dx = \underbrace{f(x, t)}_{\text{Drift term}} dt + \underbrace{\rho(x, t)}_{\text{Diffusion term}} dB_t$$

Brownian noise

$$x_{t+\Delta t} \approx x_t + \underbrace{f(x_t, t)}_{\text{Drift term}} \Delta t + \underbrace{\rho(x_t, t)}_{\text{Diffusion term}} w_t \sqrt{\Delta t}$$

$w_t \sim \mathcal{N}(0, I)$

Scales with square root so that variance increases linearly with time.

Also makes SDEs very complicated.

Backwards-in-time SDEs

$$x_{t-\Delta t} \approx x_t - f(x_t, t)\Delta t + \rho(x, t)w_t\sqrt{\Delta t}$$

**Integrating backwards in time vs
forwards in time is asymmetric!**

$$dx = f(x, t)\overset{\leftarrow}{dt} + \rho(x, t)\overset{\leftarrow}{dB}_t$$

Fokker-Planck: An SDE continuity equation

Fokker-Planck (Simplified): A time-varying family of densities $p(x_t)$ is consistent with the marginals of the **(backwards-in-time) SDE**

$$dx = f(x, t) \overleftarrow{dt} + \sqrt{2\rho(t)} \overleftarrow{dB}_t$$

For some $\rho(t) \geq 0$, $f(x, t)$ if and only if they satisfy,

$$\frac{\partial p_t(x)}{\partial t} + \nabla \cdot ([f(x_t, t) + \rho(t) \nabla \log p_t(x)] p_t(x)) = 0$$

Note: $\rho(t) = 0$ recovers the ODE transport equation!

Stochastic Sampling

We know that $p_t(x), v(x, t)$ satisfy:

$$\frac{\partial p_t(x)}{\partial t} + \nabla \cdot (v(x, t)p_t(x)) = 0$$

The SDE $dx = f(x, t)\overset{\leftarrow}{dt} + \sqrt{2\rho(t)}\overset{\leftarrow}{dB}_t$ satisfies:

$$\frac{\partial p_t(x)}{\partial t} + \nabla \cdot ([f(x, t) + \rho(t)\nabla \log p_t(x)] p_t(x)) = 0$$

We want to pick $f(x, t)$ such that the solution $p_t(x)$ is the same for both.

$$f(x, t) = v(x, t) - \rho(t)\nabla \log p_t(x)$$



The Final Steps

$$dx = f(x, t) \overset{\leftarrow}{dt} + \sqrt{2\rho(t)} \overset{\leftarrow}{dB}_t$$

$$f(x, t) = v(x, t) - \rho(t) \nabla \log p_t(x)$$

Will traverse $p_t(x)$ when using
the approximation $x_1 \sim \mathcal{N}(0, \sigma_1 I)$

We know from before that

$$v(x, t) = \dot{\sigma}_t \epsilon^*(x, t)$$

**How do we compute
 $\nabla \log p_t(x)$?**



Equivalent Parameterizations

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**x0-prediction, epsilon-prediction,
and score-prediction**



Tweedie's Formula

We have been using $\epsilon^*(x_t, t) = \mathbb{E}[\epsilon \mid x_t]$

Recall: The expectation is taken over x_0, ϵ with $x_t = x_0 + \sigma_t \epsilon$

$$\begin{aligned}\epsilon^*(x_t, t) &= \frac{1}{\sigma_t} \mathbb{E}[x_t - x_0 \mid x_t] \\ &= \frac{x_t}{\sigma_t} - \frac{1}{\sigma_t} \mathbb{E}[x_0 \mid x_t] \quad \longrightarrow \quad \epsilon^*(x_t, t) = -\sigma_t \nabla \log p(x_t)\end{aligned}$$

$$\mathbb{E}[x_0 \mid x_t] = x_t + \sigma_t^2 \nabla \log p(x_t)$$

Tweedie's Formula



Model Parameterizations

Corollary: The following are equivalent up to an affine transformation involving $x_t, \sigma_t, \dot{\sigma}_t$

$$\epsilon^*(x_t, t) = \mathbb{E}[\epsilon | x_t]$$

$$d^*(x_t, t) = \mathbb{E}[x_0 | x_t]$$

$$s^*(x_t, t) = \nabla \log p(x_t)$$

$$v(x_t, t) = \dot{\sigma}_t \mathbb{E}[\epsilon | x_t]$$

Denoising, score-matching, flow-matching, and noise-prediction models are all **equivalent**.

Back to Stochastic Sampling

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Finishing our stochastic sampler.



Denoising Diffusion Implicit Models

Continuous (Backwards) SDE

$$dx = f(x, t) \overleftarrow{dt} + \sqrt{2\rho(t)} \overleftarrow{dB}_t$$

$$f(x, t) = \left(\dot{\sigma}(t) - \frac{\rho(t)}{\sigma(t)} \right) \epsilon^*(x, t)$$

Discrete DDPM Sampler

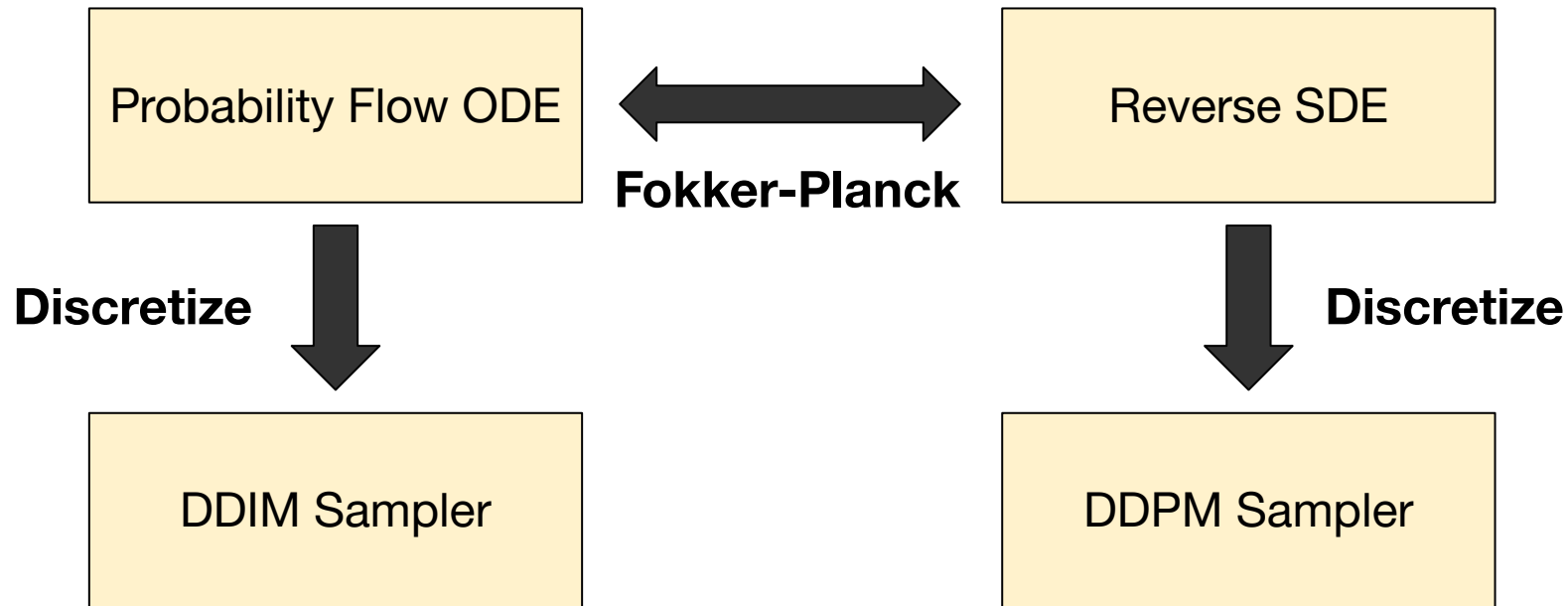
Algorithm 1 DDPM Sampling

```
1: for  $t = N, \dots, 1$  do  
2:    $z \sim \mathcal{N}(0, \mathbf{I})$   
3:    $x_{t-1} \leftarrow x_t - \left( \frac{\sigma_t^2 - \sigma_{t-1}^2}{\sigma_t} \right) \epsilon_\theta(x_t, t) + \left( \sigma_{t-1} \sqrt{1 - \frac{\sigma_{t-1}^2}{\sigma_t^2}} \right) z$   
4: end for  
5: return  $x_0$ 
```

;TLDR There is a choice of ρ , σ such that the discretization of the SDE matches the DDPM sampler



A high-level view



Recap

1. Deterministic Sampling

- a. Probability Flow ODE (Continuous)
- b. Denoising Diffusion Implicit Model (Discrete)

2. Stochastic Sampling

- a. Stochastic Differential Equation (Continuous)
- b. Denoising Diffusion Probabilistic Model (Discrete)

Fokker-Planck equation governs the evolution of a density under an SDE with the **Transport Equation** as a special case for ODEs

Many different equivalent parameterizations of diffusion models.

